ReasonMotion: Language-Guided Diffusion-Based Sports Motion Editing and Generation

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Abstract

Text-guided motion generation has seen progress, but competitive sports require a shift toward cause-and-effect modeling. In athletic training, a coach's instruction is a causal intervention; a system must not only edit a pose but simulate the downstream consequences of that adaptation. Existing supervised methods struggle with this task for two reasons. First, they focus on coarse stylistic transformations rather than the joint-level precision required for technical sports. Second, they lack biomechanical reasoning; by merely imitating ground-truth data, supervised models converge on average patterns rather than searching for optimal athletic solutions. We propose ReasonMotion, which utilizes a Spatial-Temporal-Aware Mixture-of-Experts (ST-MoE) for fine-grained instruction grounding. To enable causal reasoning, we introduce visual Group Relative Policy Optimization (vis-GRPO). Unlike imitation-based models, vis-GRPO optimizes predicted trajectories against sport-specific performance rewards, allowing the model to proactively search for high-scoring motion strategies. ReasonMotion performs well across both our sport-specific dataset FineFS and the general motion dataset Human3.6M, improving F-MPJPE by 21.4% and 12.6% respectively, shifting the paradigm from simple synthesis to consequence-aware optimization.

1 Introduction

Human motion generation plays a pivotal role in a wide range of applications, from virtual reality to sports training and robotics [37, 12, 3, 35]. With advances in multimodal learning, recent work has explored using natural language to guide human motion. This task, typically referred to as text-to-motion (T2M), is opening up new possibilities in cross-modal generation, controllable animation, and motion synthesis [4, 34, 14]. However, applying these advancements to competitive sports requires a fundamental shift in perspective. In athletic training, a coach's instruction is more than a geometric tweak—it is a causal intervention that propagates through the kinetic chain. Therefore, validating an edit requires predicting its downstream consequences along the kinetic chain, mirroring real coach–athlete feedback.

Existing T2M frameworks struggle with this cause-and-effect relationship for two reasons. First, most models focus on coarse stylistic transformations (e.g., walking to dancing) and lack the capacity for precise, localized adjustments (e.g., "bend torso forward during jump") required in technical sports. Second, these models treat motion editing and future trajectory prediction as disjointed tasks optimized via supervised imitation. Consequently, they converge on the average motion patterns of the training distribution rather than discovering optimal athletic execution. In competitive training, the goal is not to perform a likely motion, but to identify the specific kinetic adaptation that maximizes the final outcome.

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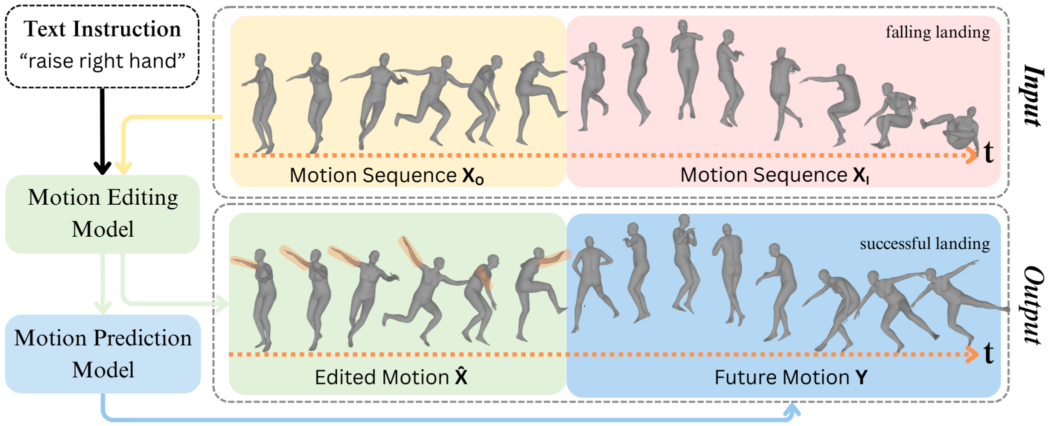

Figure 1: **An illustration of the ReasonMotion framework.** ReasonMotion acts as a biomechanical "what-if" simulator. By grounding a coaching constraint (e.g., "raise right hand") and explicitly simulating the downstream consequences, the model reasons through the kinetic chain adaptations necessary to maximize the final athletic score.

To model this complex dynamic, we propose ReasonMotion, a unified framework built on a novel biomechanical reasoning process. While large language models (LLMs) perform semantic reasoning via discrete text steps (e.g., Chain-of-Thought), ReasonMotion performs causal reasoning in the continuous physical space. It answers the fundamental coaching question: *If an athlete enters a movement with a specific technical constraint, what is the optimal way for their body to resolve the movement to achieve the highest possible score?* To achieve this, we first propose the Spatial-Temporal-Aware Mixture-of-Experts (ST-MoE), an architecture that provides the foundation for understanding by deconstructing complex motion into manageable spatio-temporal modules to accurately isolate and apply the coach's localized semantic edit.

We further introduce visual Group Relative Policy Optimization (*vis*-GRPO) to refine prediction with sport-specific rewards, shifting learning from imitation to outcome optimization. Unlike supervised imitation, *vis*-GRPO performs group-relative comparisons to drive stable policy updates. Crucially, *vis*-GRPO updates the ST-MoE routing gates, enabling the model to select expert pathways that yield higher-scoring biomechanical adaptations under the same constraint.

Our main contributions are summarized as follows:

- We propose ST-MoE, an architecture that enables structured motion decomposition and specialization through spatial and temporal experts. We empirically validate its effectiveness in modeling localized motion variations that are critical for sports performance.
- We introduce vis-GRPO, the first adaptation of the GRPO algorithm from the discrete text domain to the continuous, high-dimensional space of motion trajectory prediction.
- We propose ReasonMotion, a novel reasoning framework where the ST-MoE architecture is optimized by the vis-GRPO algorithm, enabling causal reasoning for precise motion editing and prediction tasks.
- To the best of our knowledge, this is the first work to propose a motion editing framework specifically designed for fine-grained, instruction-based adjustments in technical sports.

2 Related Work

General motion generation. Recent advances in human motion generation and prediction have introduced a variety of text- or sequence-guided models that demonstrate strong general-purpose performance [5, 32, 2]. For instance, Back to MLP [7] presents a compact MLP-based framework for 3D motion prediction; Light-T2M [31] combines Mamba and textual injection for efficient text-conditioned motion generation, while MotionDiffuse [33] employs motion VAE and classifier-free guidance to synthesize realistic and diverse motions. Despite their effectiveness on general motion, these models lack fine-grained controllability or domain adaptation essential for precise, sport-oriented motion synthesis. Other recent works such as MotionFix [1] and MotionLab [8] further enhanced controllability within motion generation frameworks, bridging the gap between free-form synthesis and targeted editing. Despite these advances, these methods remain limited in capturing semantic and biomechanical subtleties critical to athletic motions.

Sport-specific motion generation. Several works further explore sport-specific motion generation, addressing the unique biomechanical and reasoning challenges of athletic movements. Video-based methods such as ViMo [17] leverage visual cues from sports footage but are constrained to video-to-motion generation. Diffusion-based control models like SMGDiff [28] achieve real-time performance but remain domain-restricted (soccer-only), while trajectory-oriented approaches such as Sports-Traj [27] emphasize spatial coordination over joint-level precision. While these methods improve realism within sport domains, they still lack the generalizable reasoning and joint-level control required for complex athletic motion generation.

Reasoning-augmented motion generation. Reasoning-augmented motion generation leverages LLMs or structured prompting to incorporate semantic understanding into motion synthesis. MotionGPT [18] prompts low-level textual descriptions for natural motion generation but relies heavily on prompt alignment, limiting adaptability in precision-critical sports. HUMANISE [2] produces context-aware motions; however, it is limited to simple indoor scenarios. AvatarGPT [36] and Motion-Agent [25] introduce LLM-driven planning and motion-text interaction, yet they abstract away fine-grained kinematics and remain insufficiently accurate for athletic motions. Although these approaches improve contextual awareness, their "reasoning" is largely confined to the semantic text space. They are not designed for technical sports, where reasoning must involve physical, biomechanical adaptation to maximize a performance reward.

ReasonMotion uniquely bridges the gap between these domains. Unlike prior general, domain-specific, or LLM-based models, our framework specifically simulates an athlete-coach interaction.

3 Methods

The ReasonMotion architecture consists of two core components: the ST-MoE architecture to provide the decomposable structure, and the vis-GRPO algorithm to guide the model toward high-performance outcomes. The core of our reasoning process is their synergy: vis-GRPO provides the performance-driven learning signal that trains the ST-MoE's expert routing gates. This process teaches the model to reason by selecting the optimal combination of experts to maximize a reward signal, rather than just mimicking ground-truth data. The architecture of ReasonMotion is illustrated in Figure 2.

3.1 Conditional-Text Diffusion

Diffusion-based generators are effective for producing motion with rich and diverse possibilities, whereas autoregressive models often suffer from error accumulation and limited stochasticity. Motivated by these advantages, we select Temporal Cascaded Diffusion (TCD) [19], a diffusion model originally developed for motion repair and prediction, as our backbone model. We choose TCD specifically because its architecture, designed for spatio-temporal inpainting, excels at preserving the physical plausibility and continuity of existing motion. This makes it an ideal foundation for our fine-grained editing task, where the goal is to modify a specific motion part while maintaining the coherence of the rest of the sequence.

The original TCD generative process is purely coordinate-driven and therefore "blind" to text. To enable motion editing under natural-language guidance, we augment this backbone. The core TCD architecture consists of stacked ResidualBlocks, each designed to perform spatio-temporal self-attention (via time layer and feature layer modules).

We inject textual guidance by modifying each ResidualBlock. First, within the block, the motion representation X is combined with the diffusion time embedding t via addition. We then insert a cross-attention layer sequentially before the block's time layer and feature layer. For this mechanism, text embeddings from a lightweight sentence-transformer [24] serve as the Key and Value, while the

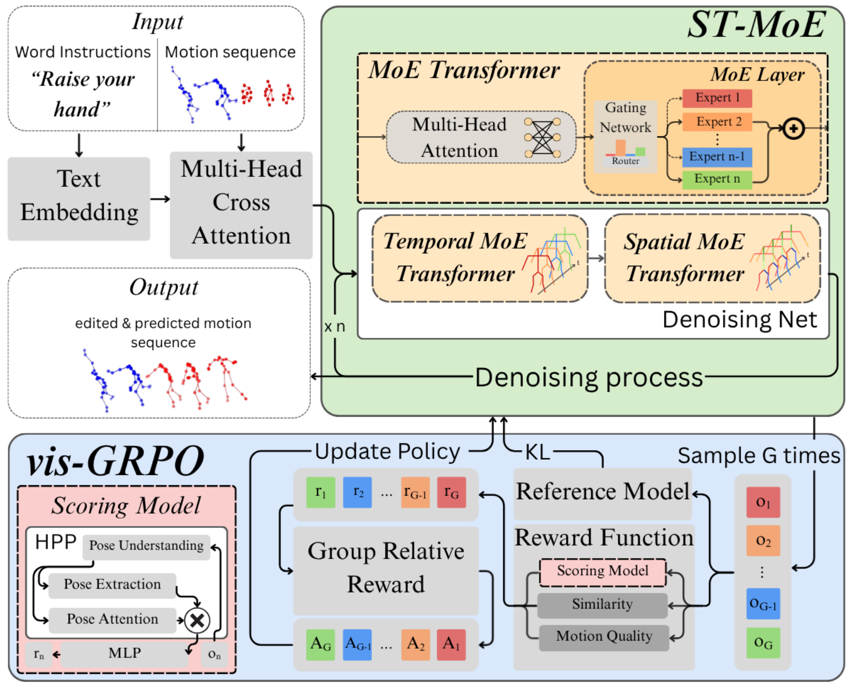

Figure 2: The ReasonMotion architecture consists of a ST-MoE diffusion model for motion generation and a vis-GRPO module for reward-guided refinement.

time-aware motion representation $(X + t)$ serves as the Query. The output of this attention layer is then added back via a residual connection, and this final text-conditioned representation is passed to the spatio-temporal self-attention layers.

Thanks to the integration of the text encoder into attention, we transform TCD into a text-conditional diffusion model that generates motion based not only on the given motion sequences but also on textual guidance.

3.2 Reasoning Process

To implement the reasoning process outlined above, we first detail the specialized policy architecture, ST-MoE. We then describe the vis-GRPO algorithm, which is the optimization framework used to train this policy with the Reward Model that provides the optimization signals for the vis-GRPO algorithm.

3.2.1 ST-MoE

As the architectural foundation of our reasoning process, we propose ST-MoE. We are inspired by the Mixture-of-Experts (MoE) paradigm [21], which has been used in models like DeepSeek-V3 [11] to enhance reasoning and reduce computational cost. However, the MoE structure is not curated for the unique demands of human motion. In motion generation and understanding, the model must have the ability to decompose a complex, continuous action into manageable, specialized parts.

ST-MoE overcomes this challenge by processing the motion representation along two distinct aspects. The **Temporal MoE** layer focuses on modeling joint-wise trajectories over time, allowing its experts to capture dynamic patterns such as acceleration, deceleration, and periodicity. In contrast, the **Spatial MoE** layer captures dependencies across joints at each frame, enabling expert specialization over body parts or coordinated structures (e.g., limbs vs. torso). Our ST-MoE is implemented within each residual block of the diffusion backbone as a sequential composition of two specialized MoE Transformer layers. Given the text-conditioned motion tensor $X \in \mathbb{R}^{B \times C \times K \times L}$ (Batch, Channels, Joints, Frames), the data flows as:

$$Y = \text{MoE-Transformer}_{\text{Time}}(X), \tag{1}$$
$$Z = \text{MoE-Transformer}_{\text{Space}}(Y). \tag{2}$$

Each MoE-Transformer is a standard Transformer Encoder Layer where the Feed-Forward Network (FFN) is replaced by a sparse MoE FFN with N expert networks and a learned router $W_r \in \mathbb{R}^{N \times d}$. Given an input token x, the router computes logits $s = W_r\, x$ and selects the Top-K experts:

$$\mathcal{I} = \text{TopK}\big(\text{softmax}(s)\big), \tag{3}$$
$$g_i = \frac{\exp(s_i)}{\sum_{j \in \mathcal{I}} \exp(s_j)}, \quad i \in \mathcal{I}, \tag{4}$$
$$\text{MoE-FFN}(x) = \sum_{i \in \mathcal{I}} g_i \cdot E_i(x), \tag{5}$$

where $E_i(\cdot)$ denotes the i-th expert network and the gate weights g_i are renormalized over only the selected experts.

3.2.2 vis-GRPO

To optimize the ST-MoE policy π_θ, we introduce vis-GRPO, a variant of the GRPO algorithm [20] adapted for visual motion generation and trajectory prediction. Current models rely on supervised learning, which restricts them to imitating ground-truth data. This approach lacks the ability to explore novel, high-scoring motion strategies crucial in sports. Our vis-GRPO reasoning process addresses this by using reinforcement learning to continuously improve motion quality based on performance rewards. Inspired by the success of GRPO in language models [6], vis-GRPO leverages relative comparisons within a generated group to provide a clear and stable learning signal. The policy optimization process is defined by:

$$J_{\text{vis-GRPO}}(\theta) = \mathbb{E}_{\substack{q \sim P(Q) \\ \{o_i\} \sim \pi_{\theta_{\text{old}}}}} \Bigg[\frac{1}{G} \sum_{i=1}^{G} \min \Bigg(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)} A_i, \\ \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\epsilon, 1+\epsilon \right) A_i \Bigg) \Bigg] - \beta D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}}) \tag{6}$$

Here, q represents the input query (e.g., observed motion and text) and $\{o_i\}_{i=1}^{G}$ is a group of G output motions sampled from the old policy $\pi_{\theta_{old}}$. The term ϵ is the clipping parameter. The advantage A_i is computed using group-wise normalization: $A_i = (r_i - \text{mean}(\mathbf{r}))/(\text{std}(\mathbf{r}) + \delta)$, where r_i is the reward for the i-th sample. Finally, $D_{\text{KL}}(\pi_\theta \,\|\, \pi_{\text{ref}})$ is a KL divergence penalty with coefficient β against a reference policy π_{ref}.

To compute the policy probabilities $\pi_\theta(o_i|q)$ in diffusion models, we interpret the policy as the stochastic reverse denoising process that induces the final output motion. Specifically, each generated motion o_i is obtained from a denoising trajectory

$$\tau_i = (x_T^i, x_{T-1}^i, \ldots, x_0^i), \qquad o_i = x_0^i, \tag{7}$$

where the trajectory probability factorizes over reverse diffusion transitions:

$$\pi_\theta(\tau_i \mid q) = p(x_T^i) \prod_{t=T}^{1} p_\theta(x_{t-1}^i \mid x_t^i, q). \tag{8}$$

Therefore, $\pi_\theta(o_i|q)$ in Eq. 6 is implemented through the likelihood of the denoising trajectory that produces o_i.

Each reverse transition is modeled as a Gaussian distribution with fixed variance:

$$p_\theta(x_{t-1}^i \mid x_t^i, q) = \mathcal{N}\left(\mu_\theta(x_t^i, t, q), \sigma_t^2 \mathbf{I}\right), \tag{9}$$

where $\mu_\theta(x_t^i, t, q)$ is the reverse-process mean induced by the diffusion model prediction, and $\sigma_t^2 \mathbf{I}$ is a fixed diagonal covariance. This fixed-variance policy provides a stable and computationally efficient signal to identify which samples are "better" than the group average, which is the core requirement for the optimization.

The primary novelty of **vis-GRPO** lies in computing policy ratios over continuous diffusion transitions. Unlike in language modeling, where the sequence probability factorizes into a product of discrete token probabilities, our policy ratio is computed over Gaussian reverse-process densities. In practice, the likelihood ratio in Eq. 6 is computed from the reverse-process transition densities. For each denoising step, we compute

$$\rho_{i,t}(\theta) = \exp\left(\log p_\theta(x_{t-1}^i \mid x_t^i, q) - \log p_{\theta_{\text{old}}}(x_{t-1}^i \mid x_t^i, q)\right). \tag{10}$$

By substituting the Gaussian transition policy in Eq. 9, the constant terms cancel out, and the step-wise log-ratio becomes a comparison of squared Euclidean distances:

$$\log \rho_{i,t}(\theta) = \frac{1}{2\sigma_t^2}\left(\|x_{t-1}^i - \mu_{\theta_{\text{old}}}(x_t^i, t, q)\|^2 - \|x_{t-1}^i - \mu_\theta(x_t^i, t, q)\|^2\right). \tag{11}$$

This formulation optimizes the policy π_θ to make its denoising transitions more likely for high-reward samples than the old policy. In implementation, rewards are assigned to the final generated motions $o_i = x_0^i$, while the GRPO likelihood ratios are computed from the per-step reverse diffusion log-probabilities along the sampled denoising trajectories.

Rationale for Fixed-Variance Gaussian Policy. Following [22] and [16], we model the diffusion reverse process as a Gaussian policy to derive a tractable log-prob ratio. While this involves a fixed-variance assumption, as demonstrated in MDM [22], this configuration is sufficient to represent human motion while maintaining mathematical completeness. Consequently, our Gaussian formulation treats the L2 distance as a valid proxy for the Evidence Lower Bound (ELBO), ensuring the vis-GRPO objective remains grounded in maximizing data log-likelihood.

Rollout Routing Replay (R3) for GRPO. Training ST-MoE with GRPO requires sampling a trajectory via a full T-step denoising chain and subsequently recomputing the forward pass with gradients enabled. In an MoE model, these two passes may yield different expert selections for the same input due to stochastic factors in training mode (e.g., dropout or router noise), causing the gradient to correspond to a computational path that was never actually traversed during rollout. This train–inference routing inconsistency has been identified as a key source of instability in MoE reinforcement learning [13].

Following the R3 framework [13], we resolve this by recording the expert assignment $\mathcal{I}_t^{\text{infer}}$ at each denoising step t during the rollout pass, and replaying it during the training pass. Concretely, at training time, the router logits $s^{\text{train}} = W_r\, x^{\text{train}}$ are still computed from the current parameters to preserve gradient flow, but the expert selection is fixed to $\mathcal{I}_t^{\text{infer}}$. The gate weights are recomputed via a *masked softmax* that restricts normalization to the recorded experts:

$$g_i^{\text{R3}} = \frac{\exp(s_i^{\text{train}})}{\sum_{j\in\mathcal{I}_t^{\text{infer}}} \exp(s_j^{\text{train}})}, \quad i \in \mathcal{I}_t^{\text{infer}} \tag{12}$$

$$\text{MoE-FFN}^{\text{R3}}(x^{\text{train}}) = \sum_{i\in\mathcal{I}_t^{\text{infer}}} g_i^{\text{R3}} E_i(x^{\text{train}}) \tag{13}$$

This formulation simultaneously achieves two goals: (1) the expert assignment is aligned between the rollout and training passes, ensuring that the policy gradient is computed on the same computational path that produced the trajectory; and (2) the gradient with respect to the router weights W_r is preserved through the masked softmax, allowing the router to continue learning.

Since the diffusion denoising chain consists of T sequential forward passes with distinct inputs x_t at each step, R3 must record and replay routing independently for each step. Thus, the rollout pass produces a routing cache $\{\mathcal{I}_t^{\text{infer}}\}_{t=T-1}^0$, and the training pass consumes one entry per denoising step:

$$\text{Rollout:} \quad x_t \xrightarrow{\mathcal{I}_t^{\text{infer}}} x_{t-1}, \qquad \text{Training:} \quad x_t \xrightarrow{\mathcal{I}_t^{\text{infer}}} \nabla_\theta \mathcal{L}_t, \quad \forall\, t \in \{T-1, \dots, 0\}. \tag{14}$$

3.2.3 Reward Model

To enable effective learning in vis-GRPO, we design a set of domain-specific reward functions r_i. These include the scoring reward r_{score}, the motion similarity reward $r_{\text{similarity}}$, and the motion quality reward r_{quality}. The total reward is computed as a weighted sum:

$$r = \lambda_{\text{score}} \cdot r_{\text{score}} + \lambda_{\text{similarity}} \cdot r_{\text{similarity}} + \lambda_{\text{quality}} \cdot r_{\text{quality}} \tag{15}$$

Scoring Reward Model. We introduce the Scoring Reward Model that assigns scalar scores to generated motions based on their perceived performance quality. This reward is crucial for guiding the model toward movements that are competitive and high-performing. We train the scoring model using labeled motion–score pairs, where each motion is annotated with a scalar score reflecting its quality (e.g., judged performance in figure skating). The Scoring Reward Model is implemented as a two-stage network. The first stage consists of a Human Pose Perception (HPP) module adapted from CoachMe [30], which encodes the spatio-temporal structure of motion sequences into compact skeletal features. The second stage is a small feedforward regression head that maps these features to a scalar score:

$$r_{\text{score}}(x) = \text{MLP}(\text{HPP}(x)) \tag{16}$$
$$= \mathbf{w}_2^\top \sigma(\mathbf{w}_1^\top \, \text{Pool}(\text{HPP}(x)) + b_1) + b_2, \tag{17}$$

where $x \in \mathbb{R}^{C\times T\times V}$ is the input motion (channels $\times$ time $\times$ joints), $\text{HPP}(\cdot)$ extracts skeletal features, and $\text{Pool}(\cdot)$ denotes spatio-temporal adaptive average pooling. The MLP head maps the pooled features to a scalar score, trained via mean squared error against the ground-truth labels. On the FineFS dataset, our Scoring Model achieves 82.13% pairwise accuracy overall, rising to 90.22% for score margins ≥ 2 (score range $[-5, +5]$), providing a precise proxy for performance that resists simple hacking.

Motion Similarity Reward. To encourage spatial and temporal consistency between the generated motion and the ground-truth motion, we define a *motion similarity reward* $r_{\text{similarity}}$ based directly on the L2 distance between their raw joint coordinates. Both motions are first flattened into vectors $\hat{x}, x \in \mathbb{R}^D$, where $D = K \times L \times 3$ corresponds to the total number of joint coordinates. The reward is then computed as the negative Euclidean distance:

$$r_{\text{similarity}} = -\left\|\hat{x} - x\right\|_2 . \tag{18}$$

This formulation penalizes dissimilar motions and encourages the model to generate outputs that closely follow the spatial and temporal structure of the reference motion.

Motion Quality Reward. To encourage low-level realism and physical plausibility in the generated motion $\hat{x} \in \mathbb{R}^{K\times L\times 3}$, we define a reward based on motion quality. Specifically, we incorporate smoothness and bone consistency rewards. The smoothness reward penalizes rapid fluctuations in velocity and acceleration:

$$r_{\text{smooth}} = -(\left\|\Delta^2 \hat{x}\right\|_2 + \left\|\Delta^3 \hat{x}\right\|_2), \tag{19}$$

where $\Delta^2 \hat{x}$ and $\Delta^3 \hat{x}$ denote discrete velocity and acceleration changes. To ensure physical consistency, the bone consistency reward penalizes temporal variation in the distances between connected joint pairs $(j_1, j_2) \in \mathcal{B}$:

$$r_{\text{bone}} = -\text{Std}_t \left(\|\hat{x}_{j_1,t} - \hat{x}_{j_2,t}\|_2\right). \tag{20}$$

The final motion quality reward is the weighted sum:

$$r_{\text{quality}} = \lambda_{\text{smooth}} \cdot r_{\text{smooth}} + \lambda_{\text{bone}} \cdot r_{\text{bone}}, \tag{21}$$

where λ_{smooth} and λ_{bone} control the contribution of each term.

3.3 Motion Prediction Model

Given an observed motion prefix $x_{\text{past}} \in \mathbb{R}^{K\times L\times 3}$ and a semantic token S (e.g., "triple lutz"), our goal is to generate a future trajectory $x_{\text{future}} \in \mathbb{R}^{K\times L'\times 3}$ that is physically coherent and consistent with the intended difficulty. This task naturally instantiates our reasoning formulation in Sections 3.2.1–3.2.3: the predictor M_P is a text-conditional diffusion policy whose ST-MoE routing gates determine *how* different spatio-temporal experts contribute to the predicted kinetic adaptation, while *vis*-GRPO refines these gates toward higher-reward outcomes.

Motion prediction requires temporal extrapolation from partial observations, and the same take-off may admit multiple plausible futures. We resolve this ambiguity by conditioning on S, which specifies the intended motion type/difficulty and thus constrains the feasible continuation.

Stage 1: SSL (Initialization). We first train M_P with the standard diffusion objective (L2 loss in the noise space) to reconstruct ground-truth x_{future} under the condition (x_{past}, S). This stage equips the model with a strong *prior* over physically plausible continuations and provides a stable initialization of ST-MoE experts and routing for downstream optimization.

Stage 2: *vis*-GRPO (Reasoning via reward-guided routing). Starting from the SSL model, we apply *vis*-GRPO to perform consequence-aware refinement. Here, M_P serves as the policy π_θ. For each query $q = (x_{\text{past}}, S)$, we sample a group of G candidate futures $\{\hat{x}^{(i)}_{\text{future}}\}_{i=1}^{G}$ from the diffusion policy (Section 3.2.2). The Reward Model (Section 3.2.3) then assigns rewards based on intrinsic motion quality (r_{quality}), sport performance proxy (r_{score}), and (when available) similarity to ground truth ($r_{\text{similarity}}$). *vis*-GRPO updates π_θ to increase the likelihood of higher-reward candidates relative to the group, effectively training the ST-MoE routing gates to select expert pathways that yield better biomechanical adaptations under the same semantic constraint.

3.4 Motion Editing Model

Given an original motion sequence $x_{\text{orig}} \in \mathbb{R}^{K\times L\times 3}$ and a text edit instruction S (e.g., "raise right hand"), our goal is to generate an edited motion $\hat{x}$ that preserves unedited regions while applying the specified modification. We address this task with a simplified TCD variant (without ST-MoE or *vis*-GRPO), adapted with two input-level modifications. First, we reduce the input from two channels to a single channel $(B, 1, K, L)$. Second, we introduce the source motion x_{orig} as an additional side-conditioning signal, appended alongside the existing auxiliary inputs [19] at each denoising step. The model denoises from pure noise under this joint condition, preventing the trivial solution of copying the input and forcing genuine instruction-driven modification. The model is trained with the standard diffusion L2 objective, as the editing task does not require the reward-guided routing refinement of *vis*-GRPO.

4 Experiments

4.1 Datasets

FineFS. FineFS [29] is a large-scale benchmark dataset for fine-grained action analysis in figure skating. It contains 1,167 video clips of professional figure skaters, including both short and free programs. Each clip is annotated with hierarchical action labels, temporal boundaries, and detailed GOE scores (-5 to $+5$). In our work, we extract skeletons using HybrIK [10], which produces 24-joint motion sequences. Skating movements exhibit strong causal relationships. For example, the way a skater prepares for a jump and the number of rotations directly impact the quality of the landing. This makes FineFS especially suitable for evaluating a model's reasoning ability and the influence of textual guidance on motion generation.

Human3.6M. Human3.6M [9] is a large-scale benchmark dataset for human motion analysis and serves as the primary dataset used to evaluate the original TCD model. It contains 3.6 million body poses spanning 15 action categories performed by seven actors. In our work, we adopt a 17-joint representation following prior work. We follow the standard [15] evaluation protocol for Human3.6M motion prediction experiments.

4.2 Experimental Design

We utilize the MotionFix dataset to train our motion editing model. MotionFix provides paired motion sequences alongside text instructions describing the edit applied between them, enabling supervised training for motion editing. The prediction task takes the source motion sequence as input alongside the corresponding text editing instruction, and the model is trained to predict the edited target motion. Specifically, for FineFS, the editing instruction is combined with the *rotation count* (e.g., "lower right hand, triple") to help the model resolve motion ambiguity and predict plausible airborne and landing dynamics. For Human3.6M, the instruction is combined with standard action-type tokens (e.g., "eating, walking") to ensure architectural consistency.

4.3 Baselines and Metrics

Our baselines include TCD, Semantic Latent Directions (SLD) [26], and Transfusion [23], a recent diffusion-based predictor demonstrating robust results on long-horizon motion prediction tasks through supervised imitation. We use this baseline to evaluate the precision of our ST-MoE in localized, fine-grained instruction grounding.

For evaluation, we report A-MPJPE and F-MPJPE (↓) for trajectory accuracy, and Diversity (↑) for motion variability. Full metric details are provided in the supplementary material.

5 Results

We compare ReasonMotion with two baseline models: SLD and Transfusion predictor (Table **??**), and our ablation variants, including TCD (base model), TCD+ST-MoE, and TCD+vis-GRPO (Table 4).

Table 1: Comparison to baselines on the MotionFix benchmark.

Model	Generated-to-Target Retrieval				Generated-to-Source Retrieval			
	R@1	R@2	R@3	AvgR	R@1	R@2	R@3	AvgR
GT	100.0	100.0	100.0	1.00	74.01	84.52	89.91	2.03
MDM-BP$_S$	38.10	48.99	54.84	6.47	60.28	69.46	73.89	4.23
MDM-BP	39.10	50.09	54.84	6.46	61.28	69.55	73.99	4.21
TMED	58.96	76.25	82.08	2.71	71.46	82.92	88.75	1.96
ReasonMotion	**73.96**	**80.21**	**86.46**	**2.47**	**55.21**	**70.83**	**76.04**	**3.82**

Table 2: Comparison to baselines on the FineFS dataset.

Model	A-MPJPE ↓	F-MPJPE ↓	GOE(mean) ↑	GOE(std)
Ground Truth	–	–	**real=0.23/pred=0.34**	**real=2.95/pred=2.68**
SLD	239.74	245.05	-3.71	1.21
TCD (Base)	218.80	236.40	0.51	0.24
Transfusion	199.83	239.68	0.19	0.12
CoMusion	162.89	187.14	0.45	0.17
ReasonMotion	**171.22**	**188.27**	**3.17**	**0.84**

In experiments on the FineFS dataset, against Transfusion, ReasonMotion outperforms it significantly, reducing A-MPJPE by 14.3% and F-MPJPE by 21.4%, while generating motion sequences with higher diversity. ReasonMotion demonstrates substantial gains in fine-grained sports motion prediction; compared to the TCD base model, ReasonMotion reduces A-MPJPE by 21.7% and F-MPJPE by 20.4%.

Most notably, when evaluating on the subset where GOE $>$ 1—representing a subsection of the dataset consisting exclusively of high-quality, successfully executed jumps—ReasonMotion shows a substantial improvement. Compared to TCD, this high-quality configuration achieves a reduction of 28.7% in A-MPJPE and 27.0% in F-MPJPE. These results prove that ReasonMotion does not merely "beautify" motion for scores; it captures the stable physical logic of elite performance, enabling exceptionally high-fidelity reconstruction of successful kinematic chains.

Results on H36M indicate that both ST-MoE and vis-GRPO achieve lower MPJPE metrics. On FineFS, our method achieves low error and high diversity, suggesting that our reasoning pipeline is optimized for complex, structured tasks rather than general motion generation.

Table 3: Comparison to baselines on the Human3.6M dataset.

Type	Model	ADE ↓	FDE ↓	MMADE ↓	MMFDE ↓
GAN-based	HP-GAN	0.858	0.867	0.847	0.858
	DeLiGAN	0.483	0.534	0.520	0.545
VAE-based	TPK	0.461	0.560	0.522	0.569
	Motron	0.375	0.488	0.509	0.539
	DSF	0.493	0.592	0.550	0.599
	DLow	0.425	0.518	0.495	0.531
	GSPS	0.389	0.496	0.476	0.525
	DivSamp	0.370	0.485	0.475	0.516
	SLD	0.348	0.436	**0.435**	0.463
DM-based	MotionDiff	0.411	0.509	0.508	0.536
	HumanMAC	0.369	0.480	0.509	0.545
	BeLFusion	0.372	0.474	0.473	0.507
	TransFusion	0.358	0.468	0.506	0.539
	CoMusion	0.350	0.458	0.494	0.506
	TCD (Base)	0.356	**0.396**	0.463	**0.445**
	ReasonMotion	**0.328**	0.427	0.470	0.498

Table 4: Ablation study of ReasonMotion on FineFS and Human3.6M.

Dataset	Model	A-MPJPE ↓	F-MPJPE ↓
FineFS	TCD (Base)	218.80	236.40
	TCD + ST-MoE	208.06	212.41
	TCD + vis-GRPO	175.99	193.81
	ReasonMotion	**171.22**	**188.27**
	ReasonMotion (GOE > 1)	**155.92**	**172.56**
H36M	TCD (Base)	112.88	146.40
	TCD + ST-MoE	110.62	128.32
	TCD + vis-GRPO	101.60	130.64
	ReasonMotion	**99.66**	**128.01**

In summary, ReasonMotion maintains competitive results on standard metrics while achieving superior motion diversity and semantic control. These characteristics make it particularly suitable for fine-grained, semantically guided motion prediction in technical sports training scenarios.

6 Discussion

6.1 Emergent Learning Characteristics

Evolution of Optimality (Fig. 3B). We analyze how the model solves predicted failures through *vis*-GRPO, shifting from probabilistic realism to biomechanical optimality. *Realism (Early Stages):* Initially, the model reflects the athlete's likely "Personal Style," accurately predicting failure from a flawed takeoff. This serves as a risk assessment, showing the probable outcome if no corrections are made. *Optimality (Late Stages):* Through RL refinement, the model discovers extreme compensatory strategies ("Textbook Form"). It learns to save failing jumps via elite mid-air positional adjustments, providing a theoretical upper bound for successful landings despite suboptimal initial conditions.

Semantic Controllability (Fig. 3C). Using a fixed takeoff (Batch 2500), we vary the prompt from "Single" to "Quadruple." The model predicts successful outcomes for 1–3 rotations but faithfully identifies that the takeoff velocity cannot support a Quadruple, resulting in a fall. This demonstrates physical honesty and robust risk assessment under constrained conditions.

Table 5: **Human Evaluation on Motion Quality and Biomechanical Plausibility.** We report the selection rate for visual realism, the percentage of trials ranked first in technical quality (GOE), physical consistency (measured by rotation count error), and instructional validity as judged by professional coaches.

Model	**Realism ↑** (Selection Rate)	**Expertise ↑** (Top-1 GOE %)	**Consistency ↓** (Rotation Error)	**Actionable Utility ↑** (Instructional Validity)
Ground Truth (GT)	13.3%	25.0%	**0.0**	N/A
TransFusion [23]	6.7%	0.0%	3.0	N/A
CoMusion [30]	0.0%	0.0%	3.0	N/A
ReasonMotion (Ours)	**73.3%**	**75.0%**	**0.0**	**50.0%**

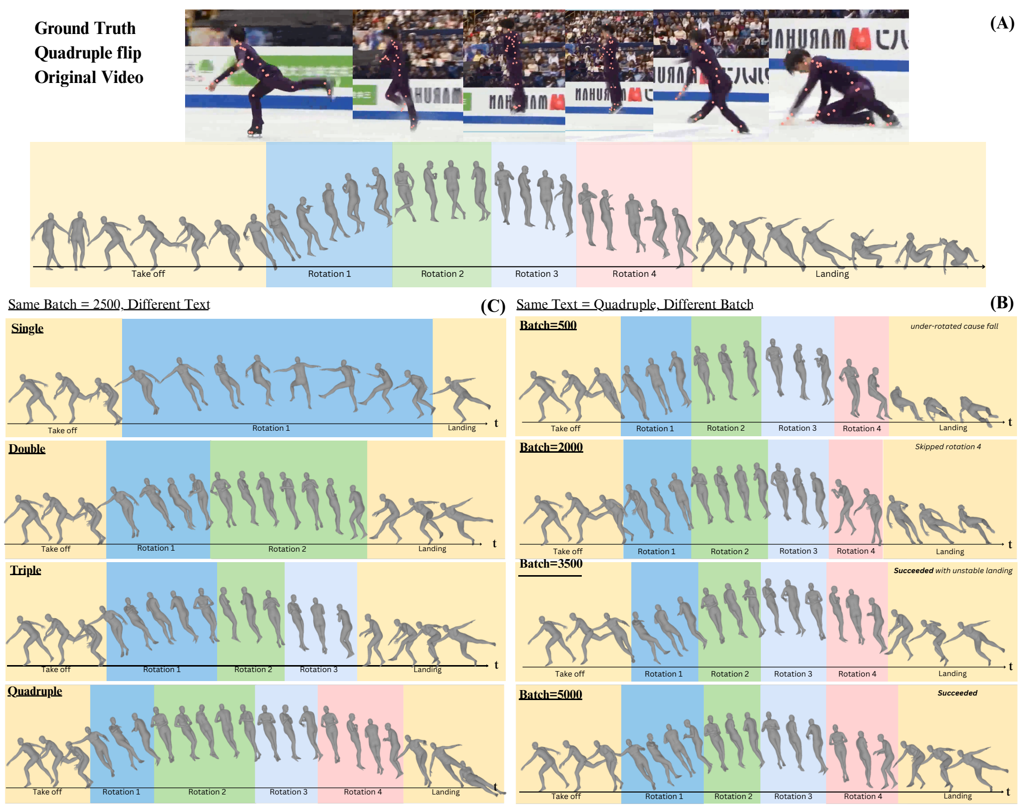

Figure 3: *Learning Evolution.* (A) Inference ground truth (a failed jump). (B) Optimal Recovery: By Batch 5000, the model discovers compensatory adjustments to "save" the landing. (C) Physical Honesty: At Batch 2500, the model correctly predicts failure (fall) for an insufficient Quadruple takeoff.

6.2 Visual Fidelity and Biomechanical Realism

To evaluate technical adjustments, we utilize an instruction-grounded module for localized edits as interventions on the takeoff (Fig. **??**). This "coaching sandbox" reveals how perturbations propagate through the kinetic chain. Unlike baselines (Transfusion, SLD) that yield "hunched" postures due to supervised dataset averaging, *vis*-GRPO prioritizes performance rewards to discover a stable, vertical kinetic chain. This biomechanical realism confirms that reward-guided reasoning produces elite-level motions, providing a reliable tool for risk assessment and strategy discovery.

7 Conclusion

This work introduces ReasonMotion, a reasoning-driven framework that elevates motion generation from stylistic synthesis to a strategic sports analysis tool. By combining ST-MoE with *vis*-GRPO, we overcome the limitations of supervised imitation, which typically converges to mediocre averages. Instead, our model proactively searches for optimal kinetic adaptations to semantic constraints. On the FineFS dataset, ReasonMotion outperforms state-of-the-art baselines with a 21.4% improvement in F-MPJPE. Beyond numerical accuracy, our model acts as a biomechanical "what-if" simulator. This shift from probabilistic realism to consequence-aware optimization empowers coaches to simulate downstream consequences of fine-grained technical adjustments, paving the way for intelligent athletic consultants.

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A Technical Appendices and Supplementary Material

A.1 Metric Details

Full details of the evaluation metrics (A-MPJPE, F-MPJPE, Diversity) are provided here.

A.2 Motion Prediction Model — Stage 2 Implementation

Detailed implementation of the vis-GRPO stage for the motion prediction model is provided here.